DIAMOND PROTOCOL
BUILT UNDER PRESSURE

Sprint positions: the angles that make athletes fast

Coach's reference · October 2026 · the positions, angles and cues behind the Sprint tab's grades

The positions that are most effective, efficient and repeatable for max velocity, acceleration and a planted cut, with the joint angles and timing behind each. Angles read as the app reads them: thigh, trunk and shin from vertical; knees as the inside angle (180 = straight). Published values are marked apart from coaching estimates in the last table.

1. Max velocity (the 30 to 40 yard clip)

At top speed the fastest athletes do not swing their legs faster; they hit the ground harder in less time, with the foot landing under the hip and already moving backward. Contact time is the best single predictor of top speed, in youth too.

Measure	Elite adults	Developing athletes (10 to 18)	What goes with speed
Ground contact	0.08 to 0.10 s (96 ± 7 ms in 26 elite men)	0.10 to 0.13 s	Shorter
Flight time	0.12 to 0.13 s	0.11 to 0.13 s	Changes little
Step rate	4.5 to 5.0 steps/s	U16 4.35, U19 4.49 steps/s	Higher
Thigh at foot strike	33 ± 4° ahead of vertical	Same shape	More in front
Thigh at toe-off	29 ± 4° behind vertical	Same shape	Less behind
Take-off knee	155 ± 7° (not locked)	Same	Smaller, not pushing to straight
Thigh angular velocity	450 to 520°/s	360 to 440°/s (estimate)	Higher, predicts foot speed at landing

Position	Look for	Common fault	Cue
Toe-off	Thighs about 90° apart; back thigh 25 to 30° behind vertical; stance knee about 155°, not locked; trunk 5 to 10° forward; free foot toe-up and tucked	Heel flicking up behind (casting); pushing to a straight knee; sitting with the hip low	"Step over the knee", "run tall"
MVP (feet level, hip highest)	Thighs still about 90° apart; front thigh at or above horizontal; front knee bent around 90 to 120° with the shin hanging vertical; back heel rising to the glute	Low front thigh; shin reaching forward early; back leg hanging	"Knee to the sky, foot under the knee"

Strike (back thigh vertical)	Front thigh 20 to 40° ahead; shin already swinging down and back; foot toe-up; swing knee closing from about 110° toward 45°	Shin flicked out ahead (overstride); toe pointing down	"Pull the ground under you"
Touchdown (first contact)	Shin vertical, within about 5°; foot close under the hip, 0.18 m ahead of the body's centre in the best elites against 0.28 m in slower ones; stance knee 150 to 160°; free knee folded to about 45° and already past the stance leg	Landing ahead of the knee (braking); heel first; knee collapsing	"Land under the hip", "stiff ankle, pop off"
Full support (hip over the foot)	Ankle under the hip, within 4 cm; no extra knee bend or ankle collapse; recovery knee already ahead of the stance knee; hip at 95% or more of standing height	Sitting; swing leg lagging behind the glute; hips drifting back	"Hips tall through the foot"

ALTIS gives the frames on purpose without angles, so the thigh angles at strike and toe-off and the take-off knee are measured values, while the MVP knee, strike separation and full-support recovery thigh are inferences from the kinograms. Front-side mechanics (less leg behind, more in front) is supported by the 2025 thigh data, but one 2017 reanalysis found the opposite sign on several of Mann's variables, so treat it as a strong coaching rule rather than a law.

2. Acceleration (the first 10 yards)

In the first steps the whole body has to sit on one forward line through the foot: the ground force points where the body leans (r = 0.93 between force direction and lean at toe-off), and trunk lean plus thigh separation at take-off explain 89% of first-step power in world-class men. Better accelerators hold the lean longer and rise toward vertical more slowly.

Measure	Step 1	Steps 2 to 5	Developing athletes
Trunk lean from vertical	45 to 50° (39° from horizontal at touchdown, 43° at toe-off in 60 m finalists)	30 to 40°	Same shape
Shin at touchdown, knee ahead of ankle	About 55°	25 to 45°	Same shape
Ground contact	0.17 to 0.19 s (0.175 ± 0.014 s world-class)	0.14 to 0.15 s	0.15 to 0.20 s
Flight time	0.04 to 0.07 s	0.06 to 0.08 s	Same
Foot at touchdown against the hip	Behind or under	Under	Same
Hip height	Low	Rising every step	Same

Position	Look for	Common fault	Cue
Toe-off	Trunk 40 to 50° forward at step 1, easing to about 30° by step 5; a straight line from ear to ankle; thighs about 90° apart; hip, knee and ankle fully extended on the push leg (knee 165 to 175°, an estimate)	Popping up early; bending at the waist with the hips back; pushing off a bent knee	"Push the ground away", "drive low and long"
MVP and strike	Short flight; front thigh punched to 60 to 70° ahead with the shin folded and the foot under the knee; back leg long with a low heel, not folded	Front knee low; heel kicking up behind	"Punch the knee, low heel"
Touchdown	Shin steeply forward, 40 to 55° at step 1 and 25 to 40° by steps 3 to 5; foot under or slightly behind the hip, within 10 cm; trunk still 30 to 45°; stance knee 140 to 150° (estimate)	Foot landing ahead of the hip and knee (the athlete stood up); heel strike; knee caving in	"Foot under the hip, shin pointing where you're going"
Full support	Trunk and shin parallel, both 35 to 45° forward; hip lower than at top speed and projecting forward; swing thigh already ahead of the stance thigh	Hip dropping below the knee line; trunk coming up while the shin stays down	"Parallel shins and chest"

A 0 to 10 yard clip usually shows steps 2 to 5 rather than the first step, so the app's acceleration references sit on the step 3 to 5 values, not the block exit.

3. Change of direction (a planted cut)

The sharper the cut, the more braking has to happen before the plant: above about 60° the penultimate step is the brake and the plant is for redirection and push. The 90° cut loads the knee most, so it is the one to coach most carefully. The fastest technique (wide plant, stiff knee, big sideways push) is also the most loading, so the app grades the safety measures first.

Position	Look for	Common fault	Cue

Penultimate step	Foot lands ahead of the hip to brake; trunk upright or leaning slightly back (0 to 10°); knee bends through 100 to 120° of range (deepest inside angle about 110 to 130°); hips, knees and ankles bend together so the body sinks before the plant	No braking here, so it all lands on the plant foot	"Brake early, sink before you plant"
Plant (first contact)	Foot pointed toward the new direction; placed wide enough to push but not wide; shin leaning into the new direction about 35 to 45° (coaching convention); knee bent at least 30° at contact (inside angle 150° or less); trunk leaning toward the new direction	Stiff straight-knee plant; trunk falling away from the cut; wide plant with the knee caving in	"Knee over the toes, nose over the knee", "lean where you're going"
Deepest point	Stance knee about 100 to 120° (peak flexion 50 to 65° in 90° cuts); knee tracking over the foot; trunk 15 to 25° toward the new direction	Knee caving in; trunk rotated away	"Knee over the foot"
Push-off	Hip, knee and ankle fully extended along the new line; shin and trunk parallel, both 40 to 45° into the new direction; first step lands under the hip	Standing up before pushing; a hopping vertical push	"Push the ground away on the new line"

The one number with injury evidence behind it is knee flexion at the plant: ACL injuries on video happen at 9 to 23° of knee flexion at contact (mean 23°), with valgus and hip internal rotation of 18 to 39° in the first 40 ms. Training the plant foot nearer the midline with a more upright trunk cut the valgus moment by 36%, and a six-week technique program in youth soccer improved cut time and reduced these faults at the same time. Plant contact times for adults, shorter being faster once braking has happened earlier: about 0.20 to 0.25 s for a 45° cut, 0.25 to 0.32 s for 90° and 0.35 to 0.45 s for a 180° turn.

4. What changes from acceleration to top speed

Measure	First steps	Top speed
Shin at touchdown	About 55° forward, then 25 to 45°	Vertical
Trunk lean	45 to 50°, then 30 to 40°	5 to 10°
Ground contact	0.18 s, then 0.14 to 0.15 s	0.08 to 0.10 s (youth 0.10 to 0.13 s)
Flight	0.04 to 0.08 s	0.12 s
Foot at touchdown	Behind or under the hip	0.2 to 0.3 m ahead of the hip
Hip height	Low, rising	Highest and steady, 95% or more of standing
Push leg at toe-off	Knee fully extended	Knee about 155°, not locked

5. The app's references

These are the starting values on Sprint → References, where the owner can change any of them. A = within ±A of the target, B = within ±B, C beyond. Bands are coaching tolerances widened for phone-video error.

Kind	Position	Measure	Target	±A	±B	Basis
Top speed	Toe-off	Thigh separation	90	12	25	Published (33° + 29° measured thighs)
Top speed	Toe-off	Stance knee	155	10	20	Published (155 ± 7°)
Top speed	Toe-off	Trunk lean	7	7	15	Estimate
Top speed	MVP	Thigh separation	90	15	30	Estimate from ALTIS
Top speed	MVP	Front knee	105	15	30	Estimate (90 to 120°)
Top speed	Strike	Back thigh	0	8	15	By definition
Top speed	Strike	Thigh separation	30	10	20	Estimate (20 to 40°)
Top speed	Touchdown	Shin angle	0	6	12	Published direction, band estimate
Top speed	Touchdown	Recovery knee	42	12	25	Published (34 to 46° elite)
Top speed	Touchdown	Stance knee	155	10	20	Estimate (150 to 160°)
Top speed	Full support	Foot under hip	0	4	9	Published direction, band estimate
Top speed	Full support	Recovery thigh	30	15	30	Estimate (must be ahead of the stance thigh)
Top speed	Step	Ground contact (s)	0.095 to 0.125 youth, 0.08 to 0.10 elite			Published
Top speed	Step	Thigh angular velocity (°/s)	360 to 440 youth, 450 to 520 elite			Estimate from range × step rate

Top speed	Step	Touchdown distance (m, lower better)	0.20 to 0.32			Published range
Acceleration (steps 2 to 5)	Toe-off	Trunk lean	35	10	20	Published (47° at step 1, falling)
Acceleration	Toe-off	Thigh separation	90	15	30	Published
Acceleration	Toe-off	Stance knee	168	10	20	Estimate
Acceleration	MVP	Front knee	95	15	30	Estimate
Acceleration	Strike	Back thigh	0	10	20	By definition
Acceleration	Touchdown	Shin angle	35	12	25	Published (55° at step 1, vertical by 30 m)
Acceleration	Touchdown	Stance knee	145	12	25	Estimate
Acceleration	Full support	Trunk lean	35	10	20	Published
Acceleration	Full support	Recovery thigh	45	12	25	Estimate
Acceleration	Step	Ground contact (s)	0.15 to 0.20 youth, 0.14 to 0.18 elite			Published
Acceleration	Step	Touchdown distance (m)	−0.10 to +0.10			Published direction
Cut (45 to 90°)	Penultimate	Stance knee, deepest	120	15	30	Published (100 to 120° of flexion range)
Cut	Penultimate	Trunk lean	5	10	20	Estimate (upright to slightly back)
Cut	Plant	Stance knee at contact	145	10	20	Published threshold (30° or more of flexion)
Cut	Plant	Shin lean	40	10	20	Estimate (coaching convention)
Cut	Plant	Trunk lean toward the new direction	15	10	20	Direction published, value estimate
Cut	Deepest	Stance knee	115	15	30	Estimate (peak flexion 50 to 65°)
Cut	Deepest	Trunk lean	20	10	20	Estimate
Cut	Push-off	Shin lean	45	10	20	Estimate
Cut	Push-off	Stance knee	165	10	20	Estimate
Cut	Step	Plant contact (s) 45 / 90 / 180°	0.20 to 0.25 / 0.25 to 0.32 / 0.35 to 0.45			Approximate, adults

6. Marking a clip in the app

1. Film side-on at hip height, the camera still, iPhone slow motion. For distances, two cones a known distance apart in the frame.
2. Pick the position, step to its frame (arrow keys move one frame; ¼ speed helps), press Use this frame.
3. Tap five points: hip, then the knee and ankle of the leg named on screen, then the knee and ankle of the other leg. The shoulder is a sixth, optional tap for trunk lean.
4. Save. The angles, each grade and the cue for anything off show at once; the rep's grade is the average of its positions.
5. Send the analysis to the athlete with a note. They see the grades, the kinogram and the cues in their app.

Sources

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Numbers come from the papers' abstracts and reported values, cross-checked against each other. Shapes transfer from adult sprinters to developing athletes; timings are loosened for youth. Where the literature gives a range or no value, the table says estimate.